EEE407 Tutorial 3b

1. Draw the International Standard Energy management system model.
2. What is included in the energy planning process conceptual diagram?
3. What are the inputs and outputs to/from management review?
4. A supermarket at Nilai has operated for 3 years. The electricity consumption for the month of March is 268,000 kWh. Every morning at 8 am, the peak electrical load reaches 1200 kW which lasted for an hour as the workers turn on all the lights and start up the centralized air-conditioning mechanical ventilation (ACMV) system, on top of the 24 hours load such as refrigeration for food products. From the total electricity consumption, the average usage for refrigeration, ACMV, and lighting is 45%, 35%, and 20%. The maximum demand declared for Connected Load Charge is 1800 kW.
   a. Calculate the electricity cost/bill based on commercial tariffs B and C1. Which tariff is lower and by how much?
   b. After an in-house energy audit, the owner decided to switch on the ACMV system earlier at 7 am – 8 am every day to reduce the peak load at peak hour. With that measure, the peak load measured at peak hour is reduced to 800 kW, while the load of ACMV during starting up is 600 kWh/day off-peak. Under this measure, calculate the possible savings that could be achieved using tariff C2 and what more could be done to further reduce the cost.
5. Table below shows the electricity consumption for a commercial complex (year 5) that used C1 tariff.

   (a) Calculate the monthly electricity bills from January until June.

   (b) Starting from July, the commercial complex has implemented Off-Peak Tariff Rider (OPTR) scheme. Compare monthly electricity bills with and without OPTR scheme from July until December.

Month	***No of days***	***Energy used (kWh)***	***Maximum demand (kW)***	***Peak energy used (kWh)***	***Off-peak energy used (kWh)***	***Maximum demand declared for CLC***
January	31	478,860	1,510	[redacted]		2100 kW
February	28	415,020	1,520			
March	31	471,760	1,540			
April	30	633,600	1,510			
May	31	527,380	1,600			
June	30	511,600	1,540			
July	31	[redacted]	1,320	485,840	121,460	
August	31		1,310	405,184	101,296	
September	30		1,280	509,440	127,360	
October	31		1,290	403,168	100,792	
November	30		1,300	410,592	102,648	
December	31		1,280	282,256	70,564	

6. A shopping mall needs to pay for Power Factor Surcharge for a particular month because of their power factor is 0.70 lagging. Calculate the cost of Power Factor Penalty they need to pay to TNB if their electricity bill for that month is RM 200,000.